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Building the Bronze Layer

"Data is a precious thing and will last longer than the systems themselves." – Tim Berners-Lee

In this chapter, you'll embark on the beginning of your data journey in the Databricks DI Platform by exploring the fundamentals of the Bronze layer of the Medallion architecture. The Bronze layer is the first step in transforming your data for downstream projects, and this chapter will focus on the Databricks features and techniques you have available for the necessary transformations. We start with an introduction to **Auto Loader**, a tool to automate data ingestion, which you can implement with or without **Delta Live Tables** (**DLT**) to insert and transform your data. The benefits of using Auto Loader include quickly transforming new data into Delta format and enforcing or evolving schemas, which are essential for maintaining consistent data delivery to machine learning models, the business, and customers. . We find that these topics are best explained through the example projects, so the section on *Applying our learning* is the main focus of this chapter.

Here is what you will learn as part of this chapter:

- Revisiting the Medallion architecture pattern
- Transforming data to Delta with Auto Loader
- Delta Live Tables starting with Bronze
- Maintaining and optimizing Delta Tables
- Applying our learning

Revisiting the Medallion architecture pattern

We introduced the Medallion architecture in *Chapter 2*. As a reminder, this refers to the data design pattern used to logically organize data transformations. A typical Medallion architecture has three layers: Bronze, Silver, and Gold. The Bronze layer contains raw data, the Silver layer contains cleaned and transformed data, and the Gold layer contains aggregated and curated data. Curated data refers to datasets selected, cleaned, and organized for specific business or modeling purposes. There are